ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** ask a question	It begins with a question.
2	**(d)** cell	All living things are made of cells.
3	**(d)** well-being of body, mind and society	WHO definition.
4	**(a)** two terminals	Positive and negative.
5	**(d)** force	Definition.
6	**(d)** force / area	Formula.
7	**(c)** particles	Tiny particles.
8	**(a)** O	O.
9	**(c)** homogeneous mixture	Uniform.
10	**(a)** reflection	Reflection.
11	**(c)** rotation	Spinning.
12	**(d)** producers	They make food.
13	**(a)** third planet from the Sun	Order of planets.
14	**(a)** variable	Variables can change.
15	**(b)** tissue	Tissue.
16	**(b)** cold	Droplets carry viruses.
17	**(c)** open circuit	No current.
18	**(c)** smooth surfaces	Smooth.
19	**(c)** eye	Calm centre.
20	**(d)** freezing	Fusion reverse.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	A quantity or factor that can change (2).	2
22	Smallest structural and functional unit of life (2).	2
23	Complete physical, mental and social well-being (2).	2
24	Cell and wires (1); Switch and bulb (1).	1+1
25	A push or a pull (2).	2
26	Force acting per unit area (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Question and hypothesis (1); Test, observe and record (1); Analyse and conclude (1).	1+1+1
28	Cell wall is present in plant cell only (1); Chloroplasts are in plant cells only (1); Plant cell has a large vacuole (1).	1+1+1
29	Clean hands (1); Safe water (1); Vaccination (1).	1+1+1
30	Cell connected to bulb by wires (1); Switch in series (1); Closed switch makes bulb glow (1).	1+1+1
31	Changes speed (1); Changes direction (1); Changes shape (1).	1+1+1
	OR: P = F/A (1); 40 / 2 (1); 20 Pa (1).	1+1+1
32	Fixed shape (1); Fixed volume (1); Cannot be compressed (1).	1+1+1
	OR: H, O (1); C, N (1); S, Ca (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Question: why does bread rise? (1); Hypothesis: yeast makes gas (1); Experiment: bread with and without yeast (1); Observe and record (1); Conclusion (1).	1+1+1+1+1
34	Cell membrane: outer covering (1); Cytoplasm: jelly-like (1); Nucleus: control (1); No cell wall (1); No chloroplast (1).	1+1+1+1+1
35	WHO definition (1); Physical health (1); Mental health (1); Social health (1); Examples (1).	1+1+1+1+1
36	Cell (1); Wires (1); Switch (1); Bulb (1); Closed circuit allows current (1).	1+1+1+1+1
	OR: Starts motion (1); Stops motion (1); Changes speed (1); Changes direction (1); Changes shape (1).	1+1+1+1+1
37	P = F/A (1); Pressure rises when area falls (1); Knife (1); Camel (1); Tractor (1).	1+1+1+1+1
	OR: Tiny particles (1); Spaces between them (1); Constant motion (1); Forces of attraction (1); Examples (1).	1+1+1+1+1
38	Composition (1); Types of atoms (1); Breaking down (1); Examples (1); Symbols/formulae (1).	1+1+1+1+1
	OR: Solute (1); Solvent (1); Sugar-water (1); CO2-water (1); Iodine-alcohol (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** iron	Iron.
2	**(c)** hot water	Heat helps.
3	**(a)** convex	Bulges.
4	**(a)** Purnima	Full moon day.
5	**(c)** food chain	Sequence.
6	**(c)** troposphere	Closest to the ground.
7	**(d)** number and unit	e.g. 5 m.
8	**(c)** cell, tissue, organ	Small to large.
9	**(c)** all nutrients in right amounts	Balanced diet.
10	**(c)** switch off the supply before fixing a plug	Safe.
11	**(c)** weight	Mass x gravity.
12	**(a)** air pressure	Atmospheric pressure.
13	**(c)** fit into the spaces	Gaps.
14	**(b)** wires	Wire drawing.
15	**(b)** mass of solute + mass of solvent	Total.
16	**(a)** focal length	f.
17	**(c)** Earth rotates	Apparent motion.
18	**(c)** deforestation	Removal of trees.
19	**(b)** rivers	Not salty.
20	**(c)** thermometer	Thermometer measures temperature.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Tiny particles (2).	2
22	Pure substance made of one kind of atom (2).	2
23	Homogeneous mixture of solute and solvent (2).	2
24	Bouncing back of light from a surface (2).	2
25	Rotation of the Earth (2).	2
26	Living and non-living things interacting in an area (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Sugar and water (1); CO2 and water (1); Iodine and alcohol (1).	1+1+1
28	Virtual (1); Erect (1); Same size (1).	1+1+1
29	Day and night (1); Apparent motion of the Sun (1); Apparent motion of stars (1).	1+1+1
30	Biotic: fish, algae, frog (1); Abiotic: water, light, temperature (1); They interact (1).	1+1+1
31	Troposphere (1); Stratosphere, mesosphere (1); Thermosphere, exosphere (1).	1+1+1
	OR: Length of a pencil 15 cm (1); Mass of a book 500 g (1); Temperature 30 degrees Celsius (1).	1+1+1
32	Human (1); Mango tree (1); Earthworm (1).	1+1+1
	OR: Bathing (1); Brushing teeth (1); Washing hands (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Statement of law 1 (1); Statement of law 2 (1); Pins and paper (1); Measure angles (1); i = r (1).	1+1+1+1+1
34	Rotation: 24 hours (1); Revolution: 365 days (1); Day and night (1); Seasons (1); Leap year (1).	1+1+1+1+1
35	Biotic (1); Abiotic (1); Producers (1); Consumers (1); Decomposers (1).	1+1+1+1+1
36	Crust (1); Mantle (1); Outer core (1); Inner core (1); Features (1).	1+1+1+1+1
	OR: Qualitative: no number (1); Quantitative: with number (1); Qualitative examples: colour, smell, taste (1); Quantitative: mass, time, length (1); Both are useful in science (1).	1+1+1+1+1
37	Scrape the cheek gently (1); Place on a slide with water (1); Stain (1); Cover with a cover slip (1); Observe and draw (1).	1+1+1+1+1
	OR: Scurvy (1); Rickets (1); Anaemia (1); Goitre (1); Night blindness (1).	1+1+1+1+1
38	Iron core (1); Coil (1); Armature and hammer (1); Gong (1); Repeating action (1).	1+1+1+1+1
	OR: Attracts iron (1); Poles attract and repel (1); Acts at a distance (1); Compass (1); Made into magnets (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** cork	Hooke observed cork.
2	**(a)** cholera	It spreads through water.
3	**(c)** positive terminal	Positive.
4	**(d)** newton	N.
5	**(a)** pascal	Pa.
6	**(b)** have spaces between them	Spaces present.
7	**(d)** H	H.
8	**(b)** solute	Dissolved substance.
9	**(b)** angle of reflection	Law.
10	**(d)** 24 hours	A day.
11	**(a)** decomposers	Break down dead matter.
12	**(d)** the Moon	One Moon.
13	**(a)** one	Only one.
14	**(c)** organ	Organ.
15	**(b)** bacteria	Salmonella.
16	**(b)** closed circuit	Current flows.
17	**(c)** friction	It's a lubricant.
18	**(b)** wind speed	Wind.
19	**(c)** sublimation	Solid to gas.
20	**(d)** sulphur	Sulphur.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Crust (1); Mantle and core (1).	1+1
22	A testable guess or explanation (2).	2
23	Contains chlorophyll (1); Helps in making food by photosynthesis (1).	1+1
24	Scurvy (1); Anaemia (1).	1+1
25	A safety device with a thin wire (1); Melts when current is excessive (1).	1+1
26	Oiling (1); Using ball bearings (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Don't touch bare wire (1); Don't use wet hands (1); Switch off before repair (1).	1+1+1
28	Stay indoors (1); Avoid trees (1); Avoid water (1).	1+1+1
29	Stay indoors (1); Listen to radio (1); Move to shelters (1).	1+1+1
30	Drying clothes (1); Drying puddles (1); Sweat drying (1).	1+1+1
31	Water (1); Salt (1); Carbon dioxide (1).	1+1+1
	OR: Solution = 50 g (1); 10/50 x 100 (1); 20 per cent (1).	1+1+1
32	n = 360/120 - 1 (1); = 3 - 1 (1); = 2 (1).	1+1+1
	OR: Natural satellite (1); Reflects sunlight (1); Shows phases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Weight of air (1); Suction cup (1); Straw (1); Syringe (1); Barometer (1).	1+1+1+1+1
34	Ice warms to 0 (1); Melts at 0 (1); Water warms to 100 (1); Boils at 100 (1); Steam warms (1).	1+1+1+1+1
35	Uniform composition (1); Non-uniform (1); Salt water, air, brass (1); Sand-water, oil-water, soil (1); Different separation methods (1).	1+1+1+1+1
36	Dissolves many solids (1); Dissolves gases (1); Dissolves liquids (1); Used in life (1); Not all substances (1).	1+1+1+1+1
	OR: Regular: mirror (1); Diffuse: paper (1); Both obey laws (1); Image only in regular (1); We see objects due to diffuse reflection (1).	1+1+1+1+1
37	Morning (1); Noon (1); Evening (1); Direction (1); Length (1).	1+1+1+1+1
	OR: Plant trees (1); 3 R's (1); Save water (1); Use public transport (1); Protect animals (1).	1+1+1+1+1
38	Lithosphere (1); Hydrosphere (1); Atmosphere (1); Biosphere (1); Links (1).	1+1+1+1+1
	OR: Independent changes by us (1); Dependent is measured (1); Constant stays same (1); Example: plant growth with water (1); Water amount is independent; height is dependent (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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